

Name: _____

Student ID: _____

Question:	1	2	3	4	5	6	7	Total
Points:	7	7	4	7	6	3	2	36
Score:								

Grade: _____

$$\text{Grade} = \frac{\# \text{total points}}{4} + 1$$

Read the instructions carefully!

- The use of a calculator, book or notes is **not** allowed.
 - Answer the questions in the spaces provided. If you run out of room for an answer, continue on the back of the page, but make sure the final answer starts on the front.
 - You can ask for extra paper to do your calculations, but you shall not submit it, the graded answers will be the ones on these papers, so make sure the final answers are properly explained here, starting in the blank space provided and finishing on the back if necessary.
 - Answers without supporting work will not be given full credit, so make sure to explain the theoretical basis of what you do for full score.
 - If you feel like you will be short in time, prioritize questions you know or the ones with higher score count.
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1. Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$ and $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ be a linear transformation given by

$$T(\mathbf{x}) = \begin{bmatrix} x_1 + 4x_2 + 5x_3 + 6x_4 + 9x_5 \\ 3x_1 - 2x_2 + x_3 + 4x_4 - x_5 \\ -x_1 - x_3 - 2x_4 - x_5 \\ 2x_1 + 3x_2 + 5x_3 + 7x_4 + 8x_5 \end{bmatrix}.$$

- (a) **(1 pts.)** Find the standard matrix A of T .

(b) (**3 pts.**) Find a basis for the Null Space of A . Is the map one-to one (injective)?

(c) (**3 pts.**) Find a basis for the Column Space of A . Is the map onto (surjective)?

2. Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & k \\ -6 & 4k & 2 \end{bmatrix}$, where $k \in \mathbb{R}$, and $b = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$.

(a) **(3 pts.)** Let $\mathbf{x} = (x_1, x_2, x_3)$. Solve the system $A\mathbf{x} = b$ in terms of k using Cramer's rule.

(b) **(1 pts.)** For which values of k is A invertible?

(c) **(3 pts.)** For $k = 1$ compute the inverse of A .

3. Let $\mathbb{P}_n$ denote the vector space of polynomials of degree at most n . Consider $U = \{p \in \mathbb{P}_n : p(2) = 0\} \subset \mathbb{P}_n$.

(a) **(3 pts.)** Show that U is a subspace of $\mathbb{P}_n$.

(b) **(1 pts.)** Let $V = \{p \in \mathbb{P}_n : p(t) = a + t^2, a \in \mathbb{R}\} \subset \mathbb{P}_n$. Is V a subspace of $\mathbb{P}_n$? Why?

4. Let

$$A = \begin{bmatrix} -2 & 2 & 1 \\ -7 & 4 & 2 \\ 5 & 0 & 0 \end{bmatrix}.$$

(a) **(2 pts.)** Find all the eigenvalues of A . Can we know from this if A is diagonalizable?

(b) **(3 pts.)** Give a basis for every eigenspace of A .

(c) **(2 pts.)** Determine whether A is diagonalizable. If so, give the diagonalization, i.e., an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. If not, explain why A is not diagonalizable.

5. Consider the following basis for $\mathbb{R}^3$: $\left\{ v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$.

(a) **(4 pts.)** Use it to find an orthogonal basis $\{u_1, u_2, u_3\}$.

(b) **(2 pts.)** Let $\mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \notin \text{Span}\{u_1, u_3\}$. Find the best approximation of $\mathbf{y}$ by vectors of $\text{Span}\{u_1, u_3\}$.

6. Mark each of the following statements as true or false. If the statement is true, give a proof. If the statement is false, give a proof or provide a counterexample.

(a) **(1 pts.)** If a matrix A is diagonalizable, then its eigenvalues are all different.

(b) **(1 pts.)** Let W be a subspace of a vector space V . If there is a vector $u \neq 0$ such that $u \in W^\perp$, then $\|u\| = 0$.

(c) **(1 pts.)** If two matrices A and B are similar, then $\det A = \det B$.

7. **(2 pts.)** A Sudoku is a 9×9 grid of numbers such that each integer between 1 and 9 occurs in every row and column exactly once. Interpreting a Sudoku as a 9×9 matrix, which can be an eigenvector of this matrix? What is the corresponding eigenvalue?

